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Gravity anomalies and crustal signature of volcano-tectonic structures of Pico Island (Azores)

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Abstract

Pico Island is the youngest (<300 ka) island of the Azores archipelago, dominated on its western sector by the stratovolcano of Pico Mountain, which rises about 3.5 km above the surrounding ocean floor. The other two major morpho-tectonic units of the island include the shield volcano of Topo, and the "Achada Plateau," a 29 km long volcanic ridge, with a WNW–ESE trend. A gravimetric study on Pico Island, coupled with geological and volcanological data, was aimed towards a better understanding of its deep volcano-tectonic structures. From a total of 407 stations, a regional model of density contrasts was developed in order to reproduce the crustal structure of the island. In addition a local high-resolution model of the sub-surface structure of the westernmost part of the island is presented. Both models were obtained by means of an inversion global adjustment, on fixed density contrasts in order to determine the three-dimensional geometry of the sources of the anomalous gravity field.

The most prominent anomalous gravity structure found in Pico Island is a body with positive density contrast located on the SE slopes of Topo volcano. This anomaly extends to depth of approximately 8 km (mean depth of 4.5 km) and has a volume of 310 km^3. This anomaly maybe best interpreted by the presence of a major intrusive system beneath Topo volcano and can be also correlated to gravitational collapses at the shield volcano. A second body of positive density contrast which appears to extend to 4–5 km depth is located near the town of Madalena. This anomaly may be correlated to a high-density anomaly recently detected in the NE sector of Faial Island. Smaller sized high-density bodies at shallow depths of around 1.3 km are located beneath Pico Mountain volcano and all together define a semi-circular arc with an average diameter of 8 km. We propose that this arcuate structure might outline a ring fracture associated with the collapse of the central sector of Pico Mountain volcano early in its growth.

Low-density bodies are located on the NW flanks of Pico Mountain, along a linear ENE trending structure. These bodies extend to depths of 4 km and have a total volume of 69 km^3. They are interpreted as being associated with an "old" extensional fracture zone. Previous studies suggested the existence of a shallow and large magma reservoir (with volumes up to 1000 km^3) beneath Pico Mountain volcano. In this study we have not found any evidence of the existence of such huge magma body. On the contrary, we propose that those discrete low-density bodies correspond to complex intersections of different sets of tectonic lineaments, along which magma tends to intrude, as shallow (2–5 km) and small (10–20 km^3) magma pockets.
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Keywords: Pico Island; gravity measurements; deep structures; magma reservoir; active volcano

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1. Introduction

Oceanic islands like those of the Azores archipelago are volcanic edifices that typically evolve over millions of years (Malengreau et al., 1999), only a small part of their eruptive history being accessible through outcrops. Therefore, geophysical data provide a valuable approach to gain a more complete picture of volcanic edifices and sub-surface structures, including the presence of magma reservoirs (e.g. Rousset et al., 1989). Gravity studies in volcanic areas in particular can provide unique insights into shallow sub-surface density variations associated with the structural and magmatic history of an active volcanic system (Rymer and Brown, 1986).

The archipelago of the Azores Islands is located in the North Atlantic Ocean, about 1600 km from Portugal mainland (Fig. 1). Composed of nine inhabited islands and several islets of volcanic origin, the WNW–ESE-trending archipelago covers an area about 600 km long and 200 km wide. The Azores islands can be divided into three groups. The western group (Flores and Corvo islands) is located west of the Mid-Atlantic Ridge (MAR), while the central (Terceira, Graciosa, São Jorge, Pico and Faial Islands) and the eastern groups (São Miguel, Santa Maria islands and Formigas islets) are located east of the MAR.

A regional gravimetric study based on a large marine data set over the entire archipelago, identified two large gravity anomalies (Luis et al., 1998): the first one trends perpendicularly to the Azores archipelago axis and the second, more discrete, is perpendicular to the Mid-Atlantic Ridge. The study also demonstrated that the gravity data from the Azores Plateau area maybe successfully modelled via elastic plate models. These models gave values of 12–14 km for the depth of the crust–mantle interface (or Moho), representing an average crustal thickness of about 11 km, using a mean sea floor depth of 2 km.

Pico Island is the youngest island of the Azores archipelago, dominated on its western sector by Pico Mountain, an impressive 2351 m high (a.s.l) stratovolcano. To better define the deep volcano-tectonic structures of Pico Island and to refine our understanding of their local and regional implications, gravimetric studies were carried out on Pico Island, as a joint cooperation program of "Instituto de Astronomía y Geodesia" (CSIC–UCM, Madrid) and the "Universidade dos Açores" (UA, Azores). Similar studies have also been conducted on São Miguel Island (Camacho et al., 1997), Terceira Island (Montesinos et al., 2003) and Faial Island (Camacho and Vieira, 2002). The purpose of this paper is to present and interpret gravimetric data recently obtained for Pico Island and to correlate them

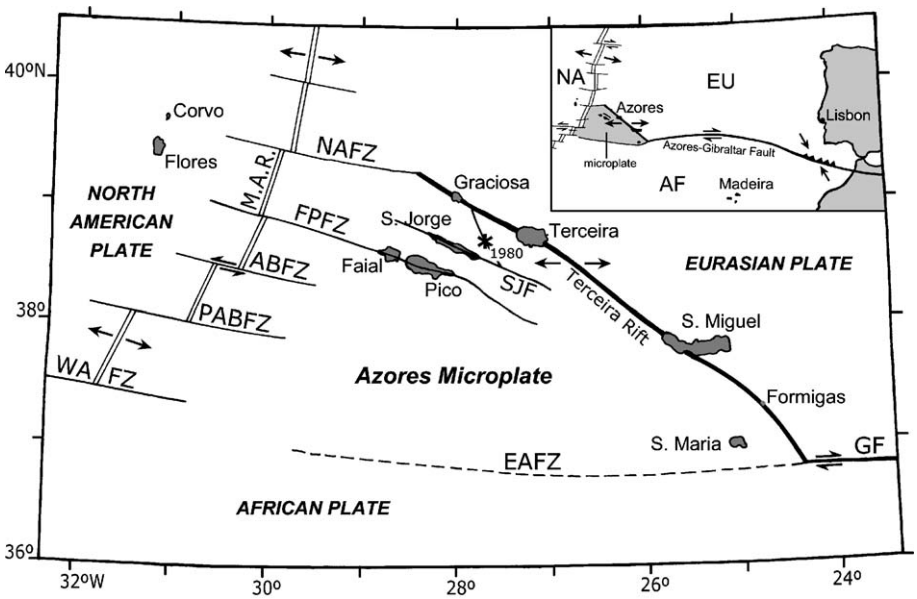

Fig. 1. General geotectonic framework of the Azores archipelago (adapted from Montesinos et al., 2003). A star indicates the location of the 1980 January 1st earthquake. MAR = Mid-Atlantic Ridge; EAFZ = East Azores Fracture Zone; WAFZ = West Azores Fracture Zone; NAFZ = North Azores Fracture Zone; FPFZ = Faial–Pico Fracture Zone; ABFZ = Açor Bank Fracture Zone; PABFZ = Princess Alice Bank Fracture Zone; GF = GLORIA Fault; SJF = São Jorge Fault; EU = Eurasian Plate; AF = African Plate; NA = North American Plate.

with morpho-tectonic and volcanic features identified during fieldwork. We also expect these data to contribute to a better assessment of volcanic hazards on Pico Island.

2. Geodynamic framework of the Azores triple junction

Azores Islands rise from an anomalously shallow and rugged topographic zone (the so-called "Azores Plateau"—Needham and Francheteau, 1974), crudely triangular in shape and having depths less than 2 km. This plateau occupies an area of 5.8×10^6 km^2 and marks the transition to the adjacent abyssal plains, with depths exceeding 3.5 km.

The Azores region contains the triple junction between the Eurasian, African and North American Plates (Searle, 1980; Forjaz, 1983; Luis et al., 1994; Lourenço et al., 1998). The islands are aligned along major tectonic lineaments with a general WNW–ESE trend, defining the so-called Azores "microplate" or "block," an area of active volcanism and high seismicity. The location and nature of the northern and south-western arms of the Azores triple junction are well established and correspond to the Mid-Atlantic Ridge (MAR), a pure extensional structure which is placed between Flores and Faial Islands, representing the boundary between the North American Plate and the two other plates (Fig. 1).

The nature and precise location of the third branch (generally following a NW–SE trend) is still under debate, especially in the area of the Azores islands, despite the several kinematic models proposed for the Azores triple junction (e.g., Krause and Watkins, 1970; Machado et al., 1972; Searle, 1980; Forjaz, 1983; Buforn et al., 1988; Madeira, 1998). In general terms however, the third arm of that RRF (Ridge–Ridge–Fracture) divergent triple junction corresponds to the Azores–Gibraltar Plate fault zone (Ribeiro, 2002).

This boundary among the Eurasian and African plates outlines three sectors with distinct geodynamic behaviour (Fig. 1): (i) an eastern sector (east of 15°W), with overthrusting of the Eurasian plate over the African plate; (ii) a central sector (west of 15°W and 800 km long) that corresponds to a well-defined E–W right lateral-slip fault (e.g., GLORIA Fault — GF); and (iii) a WNW–ESE-trending branch, extending westwards from Santa Maria Island. In global kinematic models, this western branch shows a right-lateral transtensional regime (Madeira, 1998), with an extensional component of 4 mm/yr full spreading rate (Luis et al., 1998) and a dextral movement on the order of 11 mm/yr (Ribeiro, 2002). According to Luis et al. (1994), the Faial–Pico Fracture Zone at present exhibits high seismicity, has clear bathymetric and magnetic expressions and probably defines, at its intersection with the MAR, the present position of the Azores triple point.

The regional tectonic lineaments and local fault systems crossing the islands are indicated by significant seismic and volcanic activity, observed in historical times on the Central Group islands and São Miguel. The most recent volcanic eruption ("Serreta eruption") was submarine, West of Terceira Island and lasted from late 1998 to early 2000. The most recent moderate to strong earthquake occurred in the Azores on 9 July 1998 (off-shore Faial Island), with a magnitude (M_d) of 6 and intensity VIII/IX on the Modified Mercalli Intensity Scale.

3. Pico Island geology

Pico Island is located on the Faial–Pico Fracture Zone (Fig. 1), a 350-km long leaky transform structure that extends from the MAR towards São Miguel Island. The elongated shape of Pico Island along this tectonic alignment reflects the strong tectonic control of its volcanism, which is mostly Late Pleistocene in age. Maximum known radiometric ages for Pico (300 ka — Chovelon, 1982), Faial (800 ka) and São Jorge islands (600 ka — Feraud et al., 1980) show a generally young age of their subaerial parts, compared to the remainder of the Azores islands (Feraud et al., 1980; Abdel-Monem et al., 1975). However, these ages do not indicate any clear migration pattern, or shift, in the volcanic activity in this area of the Azores Plateau, namely in relation to the Mid-Atlantic Ridge (130 to 180 km to the west). Also, the radiometric ages are not compatible with the mechanism of simple movement of the lithospheric plate over a small and fixed hot spot (Feraud et al., 1980).

Three major morpho-tectonic units can be recognized for Pico Island (Fig. 2): (i) a high stratovolcano, Pico Mountain, that dominates the western part of the island; (ii) a shield-like volcanic structure, Topo volcano, located on the south-central part of the island; and (iii) a 29-km long, WNW–ESE volcanic ridge, the "Planalto da Achada," comprising the eastern part of the island.

The basaltic stratovolcano of Pico Mountain rises 3.5 km above the surrounding ocean floor. Its subaerial volume is 97 km^3, as compared to the total subaerial volume of Pico Island of 207 km^3 (Nunes, 1999). This volcano's slopes are gentle (5° or less) up to an elevation of 200 m, sharply increasing to about 30–35° above 1200 m and locally reaching as much as 60°, in the

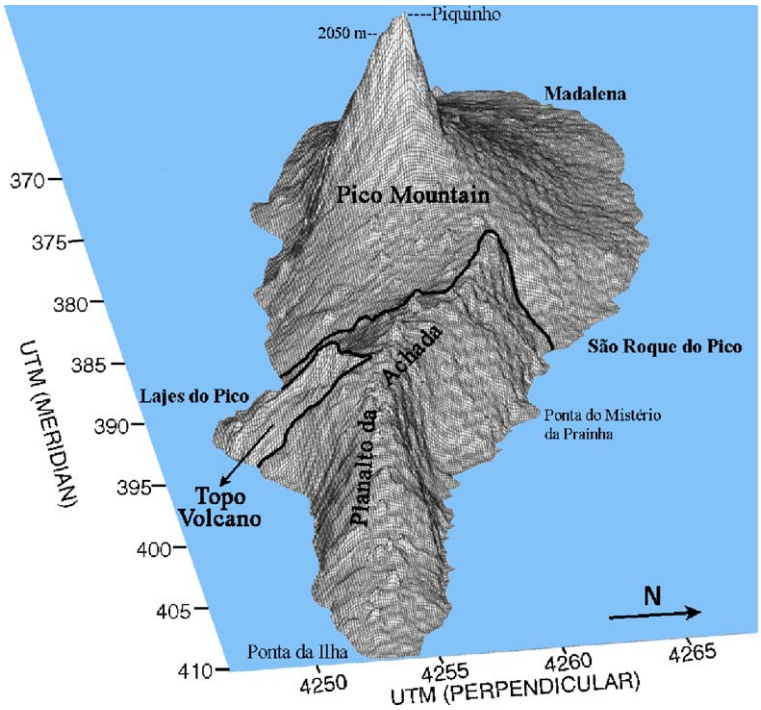

Fig. 2. Digital elevation model (DEM) for Pico Island. Perspective from the east (adapted from Nunes, 1999), with indication of the main morphotectonic units of the island (Pico Mountain stratovolcano, Topo shield volcano and "Planalto da Achada" volcanic ridge). UTM coordinates in km.

summit area. At 2250 m high, a collapse crater, 550 m wide and 25 m deep, occupies the top of the mountain. The crater bottom is covered with *pahoehoe* lavas emerging from a nested, 125-m high driblet cone, Piquinho. The rim of an older crater (800 m in diameter) is visible at 2050 m on the southern slope of the volcano (Fig. 2).

Several flank eruptions have occurred, mainly on the western and southeastern flanks of Pico Mountain, producing scoria or spatter cones and eruptive fissures. Extensive lava flows have completely covered the base of the volcano; eruptive vents for many of the flows are located at higher levels of the volcano. Based on the volcanic productivity for Pico Island over historic eruptions, the maximum expected age for the subaerial part of the volcano is assumed to be around 240 ka (Nunes et al., 1999). The volcanic history of Pico Mountain indicates significant activity during the Holocene, for example the 1700 and 1300 years BP lavas, erupted from the present summit crater and from the summit Piquinho cone, respectively (Fig. 2). Since settlement, in the 15th century, Pico Mountain erupted occasionally in the period 1718–1720.

Topo volcano rises 1022 m (a.s.l.) and 2500 m above the ocean floor. It is a basaltic shield volcano built of fluid lava flows with 5–10% pyroclastic material. Its volume above sea level is 8 km^3, and its present exposed surface area is 18 km^2. Several tectonic escarpments, with general N–S to NE–SW trends, dissect the volcano. It has two subsidence structures (Terras Chãs and Santa Bárbara), which some authors have interpreted as volcanic calderas (Fig. 3). While Santa Bárbara subsidence area may perhaps represent an old collapse crater (or caldera), Terras Chãs depression is best interpreted as having formed by a lateral movement of the flank of the volcano (Nunes, 1999, 2003), driven by gravity and tectonic instability, rather than by caldera collapse at the summit of the volcano. The Terras Chãs depression is 3.5 km long, 1–1.5 km wide, and open to the sea on the south; it is bordered on the west by the Arrife Fault and related structures. The oldest subaerial rocks of Topo volcano has been dated at 250 ka (Chovelon, 1982), and the youngest lavas extend into the Holocene.

The "Planalto da Achada" (or Achada Plateau) is composed of numerous volcanic cones located along the WNW–ESE-trending axis of the plateau. Lavas issuing from these cones were mostly *aa*-flows that cascaded down the steep northern or southern slopes of the plateau. 170 volcanic edifices have been identified on this ridge, mostly scoria cones and including small

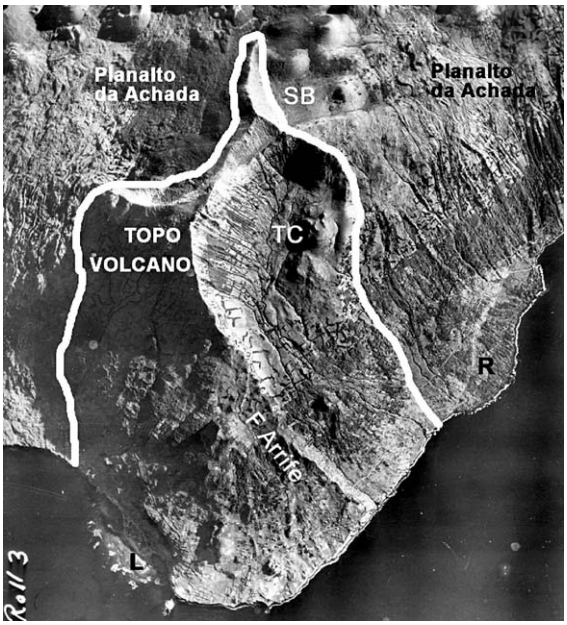

Fig. 3. Topo volcano aerial view, showing Terras Chãs (TC) and Santa Bárbara (SB) depressions, the Arrife Fault scarp (F Arrife) and the lava deltas of Ribeiras (R) and Lajes do Pico (L). Aerial photo by USAF.

cones associated with eruptive fissures. The oldest subaerial formation has an age of 230 ka (Chovelon, 1982), and the youngest volcanic event was the 1562–1564 A. D. eruption of Mistério da Praínha. Volcanic activity has shifted variably between different parts of the Achada Plateau for short periods of time, but there is no indication of a clear temporal or spatial migration pattern of the eruptive vents along that ridge.

Rocks from Pico Island are mostly basalts and hawaiites, falling within a narrow part of the alkaline series compositional field without any strongly evolved components. A single flow with mugearitic to benmoreitic composition was identified in the 1718 AD lava at "Mistério de Santa Luzia" (França et al., 1995). Rocks from Pico Island display a wide variety of textures, from aphyric lavas to ankaramites, and porphyritic lavas. Scoria deposits are restricted to the proximities of volcanic cones, except for the higher levels of Topo volcano and on the Achada Plateau axis, where basaltic ash and lapilli layers mantle the surroundings areas.

4. Seismo-tectonics of Pico Island

Available data clearly show that historical volcanic eruptions were preceded and followed by notable seismic activity, which sometimes caused damage in the nearby areas. However, earthquakes that shake Pico Island are fewer and lower in magnitude than on the other islands of the central group; the most seismically active areas are Pico Mountain and the adjoining Faial–Pico channel between the two islands. About 90% of felt earthquakes have intensities less than IV and do not cause any damages (Nunes, 1999). However, Pico Island is occasionally hit by strong earthquakes with intensity higher than VI/VII, usually with epicentre in the nearby seismic zones (e.g. the 1757, 1926 and 1998 earthquakes).

Three major groups of tectonic lineaments can be seen on Pico Island (Fig. 4): (i) WNW–ESE lineaments, most common and directly controlled by the regional stress field; (ii) NNW–SSE structures, mostly as left-lateral strike-slip faults with a dip-slip component (Madeira, 1998), like the "Lomba do Fogo" eruptive fissure, more than 25 km long and focus of the 1718 A. D. eruption; and (iii) NE–SW to ENE–WSE tectonic lineaments, which are dominant on the central polygenetic volcanoes of Pico Mountain and Topo, where they seem to correspond to radial fractures. Some of the latter structures have a dip and a slight slip component; this trend is also common for most of the dikes found in the eastern half of the island.

5. Gravity data and Bouguer anomaly

Gravity measurements were carried out on the island in July 1997, July 1998 and April 2002, using two LaCoste and Romberg gravimeters (G933 and G665), each with electronic reading system. A total of 506 measurements were made at 407 stations (see Fig. 5 for station distribution), mostly along roads. On average, four readings where taken at each station, and repeated observations in several stations and in different days, allowed us to control the behavior of the gravimeter (e.g. drifting).

A nearly homogenous coverage of the island, with an average distance of 1.5 km between contiguous data points, was obtained except for the higher slopes of the Pico Mountain. The westernmost part of the island has denser coverage, with average distances of about 1.1 km between data points, as the result of the 2002 fieldwork, which focused on a better definition of local and regional tectonic features related to the Faial–Pico Fracture Zone (see also Section 7). For gravity data correction, barometric (or also GPS) elevations at each gravity station have been obtained, employing known geodetic data as control values. A mean precision of 2 m was estimated for the elevation of each station where barometric levelling was done.

By means of using repeated observations (16%), we were able to apply an adjustment to the entire network to

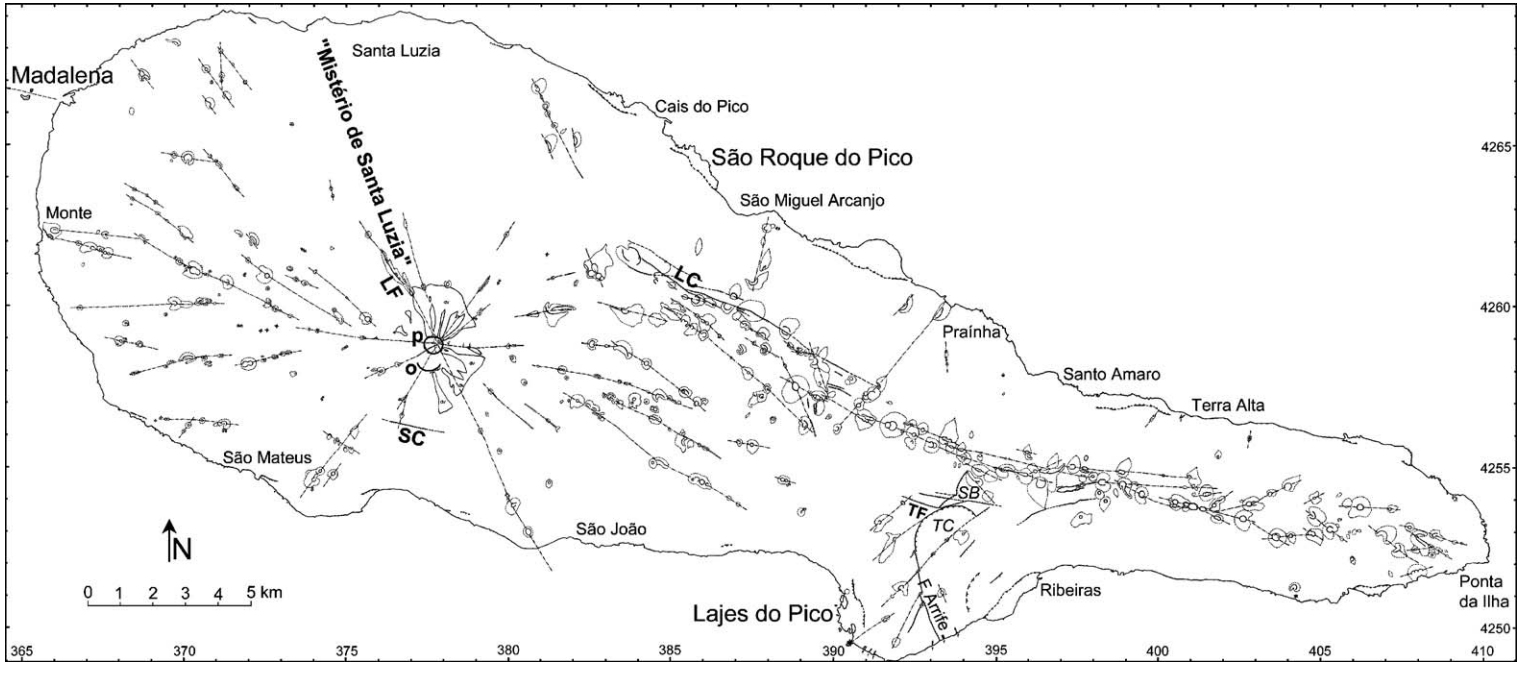

Fig. 4. Morpho-tectonic map of Pico Island (Nunes, 1999). LF = Lomba do Fogo fault; LC = Lagoa do Capitão fault; SC = São Caetano fault; TF = Topo fault; TC = Terras Chãs depression; SC = Santa Bárbara crater; FArrife = Arrife fault scarp; p = Pico Mountain summit pit crater; o = Pico Mountain older crater, at 2050 m elevation. UTM coordinates in km.

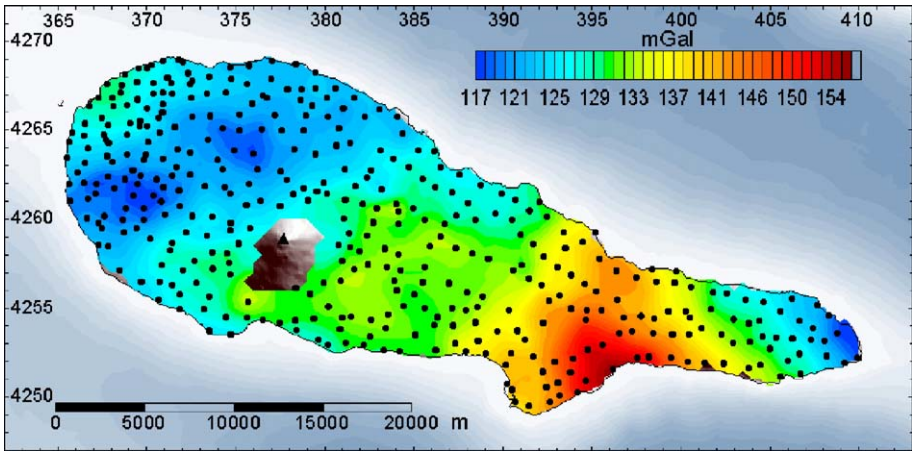

Fig. 5. Bouguer anomaly map derived from gravity data reduction for a terrain density of 2520 kg/m^3 (see text for details). Dots show the location of the 407 gravity stations occupied during the surveys. The triangle corresponds to Pico Mountain summit. UTM coordinates in km.

determine the drifts of the gravimeters and a relative scale factor. The root mean squared value for the residual values of the gravity observations (corrected for tidal effects) is 0.022 mGal (1 mGal = 10^{-5} m s^{-2}). Subsequently, our gravity data were referred by direct connection to the absolute gravity value of 980,129,217 ± 7 mGal determined in 1997 with the GILAg-5 of the Finnish Geodetic Institute (Bastos et al., 2000) at "Observatório Meteorológico Alberto do Mónaco," in Horta, Faial Island.

Taking into account the strong relief of the island, the terrain correction for the gravity data is significant in the reduction process. Therefore, a digital elevation model (DEM) was constructed (see Fig. 2), with a 125 m horizontal resolution based on 14,200 readings from local topographic charts (on scale 1:25,000) and from 2000 bathymetric readings up to distances of 40 km from the island.

To calculate the terrain correction for the gravity data, a mean terrain density must be selected. This density value is of critical importance for the generation of Bouguer anomaly maps and traditionally accounts for the largest uncertainty during static gravity surveys. It is therefore essential to base the selection of the terrain density on constraints obtained for example from numerical considerations or direct measurements. Demande et al. (1982), using a small number of gravity data on Pico, suggested a terrain density value of 2200 kg/m^3, while earlier Machado (1956) had assigned a mean density less than 1600 kg/m^3 for the upper levels of the Pico Mountain. This low value allowed him to explain the impressive slopes of the Pico Mountain volcano (sometimes higher than 45°), assuming the existence of a huge scoria cone on the higher parts of the volcanic edifice buried by young lava flows. However, this assumption is not supported by recent geological data (Nunes, 1999).

On the other hand, the application of Nettleton's (1939) method of minimum correlation between elevation and gravity anomaly suggests a value of 2650 kg/m^3 for the Pico Island. However, Nettleton's method is effective only if the density of the geological formations is homogeneous and if there is no inherent correlation between Bouguer anomalies and topography (Malengreau et al., 1999). In volcanic areas, Bouguer anomalies are often related to topography, because the highest areas usually correspond to eruptive centres where denser rocks are emplaced at depth, as dykes or sills. Therefore, the terrain density value used in subsequent computations was obtained from the gravity data, applying the method proposed by Camacho et al. (2002). This method looks for the least distortions in the Bouguer anomaly map, with such distortions typically related to non-existent anomalous structures within the topographical masses. As expected with this methodology, we obtained a more reasonable density value, which represents the entire topographic mass of the island.

Our derived density value of 2520 kg/m^3 is slightly lower than that obtained from Nettleton's method. Moreover, our value is consistent with several experimental density determinations obtained for basaltic lava flows on São Jorge Island, which are lithologically similar to those of Pico Island. Those determinations gave mean density values of about 2610 kg/m^3, ranging from 2250 kg/m^3 for vesiculated lava flows to 2840 kg/m^3, for dense and massive rocks (Motta and Nunes, 2003). An average density of 1307 kg/m^3 was determined for the São Jorge basaltic scoria deposits.

Additionally, for the mugearitic/benmoreitic Pico lavas of the 1718 eruption (Santa Luzia) a density of 2611 kg/m^3 was measured experimentally, at "Laboratório Regional de Engenharia Civil" (Azores).

The resultant terrain correction for the gravity data ranges between 3.2 and 30.7 mGal. The resulting Bouguer anomaly map (Fig. 5), shows a gravity anomaly that oscillates around a mean value of 129 mGal, with a standard deviation of 7.5 mGal. In Fig. 5, a clear maxima–minima distribution is observed. Using a covariance analysis (Moritz, 1980) for the horizontal relation among the anomaly data, we estimated a mean observational noise of about 0.7 mGal for the obtained gravity–anomaly distribution. It should be emphasized that the gravity anomalies pattern obtained exhibits a regional trend, with gravity decreasing from the SE to the NW (roughly along N43W), with a magnitude of about 0.3 mGal/km. This overall decrease can be associated with the presence of the Mid-Atlantic Ridge gravity minima to the W of Pico Island.

Fig. 6 presents the Bouguer anomaly map obtained for the westernmost part of the island, with a smoother topography and where a more detailed survey was planned. Therefore, this area has been investigated via a denser station network. For the terrain correction a mean density of 2580 kg/m^3 was obtained by employing the same methodology used before.

6. Gravity inversion model

The Bouguer anomaly map (Fig. 5) allows inferring on the sub-surface mass variation of the island, but

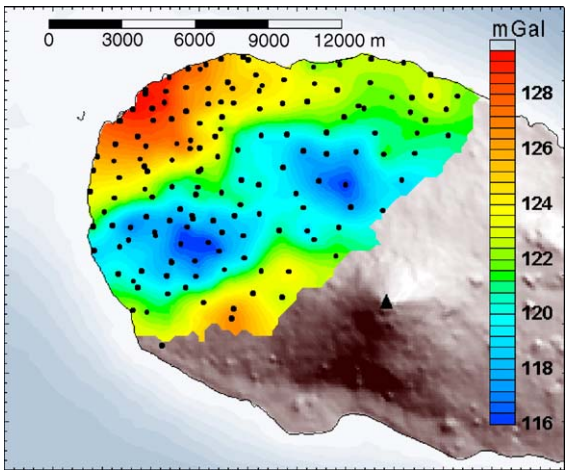

Fig. 6. Local Bouguer anomaly map for the western sector of Pico Island obtained from a total of 154 gravity stations. Gravity reduction using a terrain density of 2580 kg/m^3. The triangle corresponds to Pico Mountain summit.

additional information can be gained from a gravity inversion of the anomaly, which allows us to locate and model the sub-surface mass distribution. The non-uniqueness of gravity inversions is well known, but by formulating additional constraints to the inversion from geological observations and mathematical boundary conditions, realistic models can be obtained to infer the sub-surface anomalous bodies. Even better results are obtainable using non-linear approaches, to determine the geometry of the causative bodies, adopting a priori values for their anomalous density contrasts as outlined below. Perhaps the best way to treat these non-linear problems is the explorative approach, with random or systematic exploration of the model space. In Camacho et al. (2000), a gravity inversion method by means of a growth process of the causative bodies is described. Here, an exploratory approach allows the construction of the anomalous bodies as step-by-step aggregation of prismatic cells. The non-uniqueness problem is avoided by including a mathematical boundary condition in the form:

$$\left(\sum \text{weighted residuals of the inversion}\right) + \lambda\left(\sum \text{weighted anomalous mass}\right) = \text{minimum}$$

The first term extends along the gravity points, and the weighting or covariance matrix comes for the estimated data confidence. The second term extends along the sub-surface cells of the model, and the weighting system is defined by a suitable covariance matrix corresponding to the design of the mathematical problem. The positive parameter λ checks the balance between model fitness (residuals) and model smoothness (involved mass amount and complexity). Details and the corresponding software package of this method are presented in Camacho et al. (2002).

Using the gravity anomaly data obtained for Pico and applying the former gravity inversion method, a three-dimensional (3-D) model for the location and morphology of the several anomalous bodies under the island was calculated. The mathematical boundaries for the density contrasts range between the extreme values of −300 and 300 kg/m^3, using a mean contrast value of −254 kg/m^3, for the mass deficiency, and 256 kg/m^3, for the mass excess. These values are selected empirically, taking into account that the calculated model must plausibly correspond to subsurface bodies compatible with available geological information. Nevertheless, the choice of these values is not a critical question, because

even using different density contrasts, the resulting solutions are rather similar; however, for higher contrast, smaller volumes would be obtained.

The 3-D model obtained from our method is composed of about 8700 cells, each with a mean side length of 0.8 km, with partition of the sub-surface volume of the island from the surface to a depth of 9 km. About 2500 of these cells are assigned with the prescribed anomalous density contrast in determining the anomalous bodies. This structural model is shown in Figs. 7 and 8. The root mean squared value for the residuals of the inversion process is about 0.5 mGal.

Fig. 9 shows the model derived for the westernmost part of Pico Island, using an extreme density contrast of −300 and 250 kg/m^3 and smaller cells. The resulting mean value for the residuals is now reduced to about 0.2 mGal.

7. Results

The sub-surface model for Pico Island can be visualized by means of horizontal sections and vertical profiles (Figs. 7 and 8 for the whole island, and Fig. 9 for its westernmost part). Despite the limitations of the solution of the gravity inversion we can infer on the following (see also Figs. 2–4 for names and locations):

1. The high-density bodies of the island form three main groups: (A) a large body located in the SE zone of Pico Island, near Ribeiras and Terras Chãs depression; (B) a group of shallow structures scattered

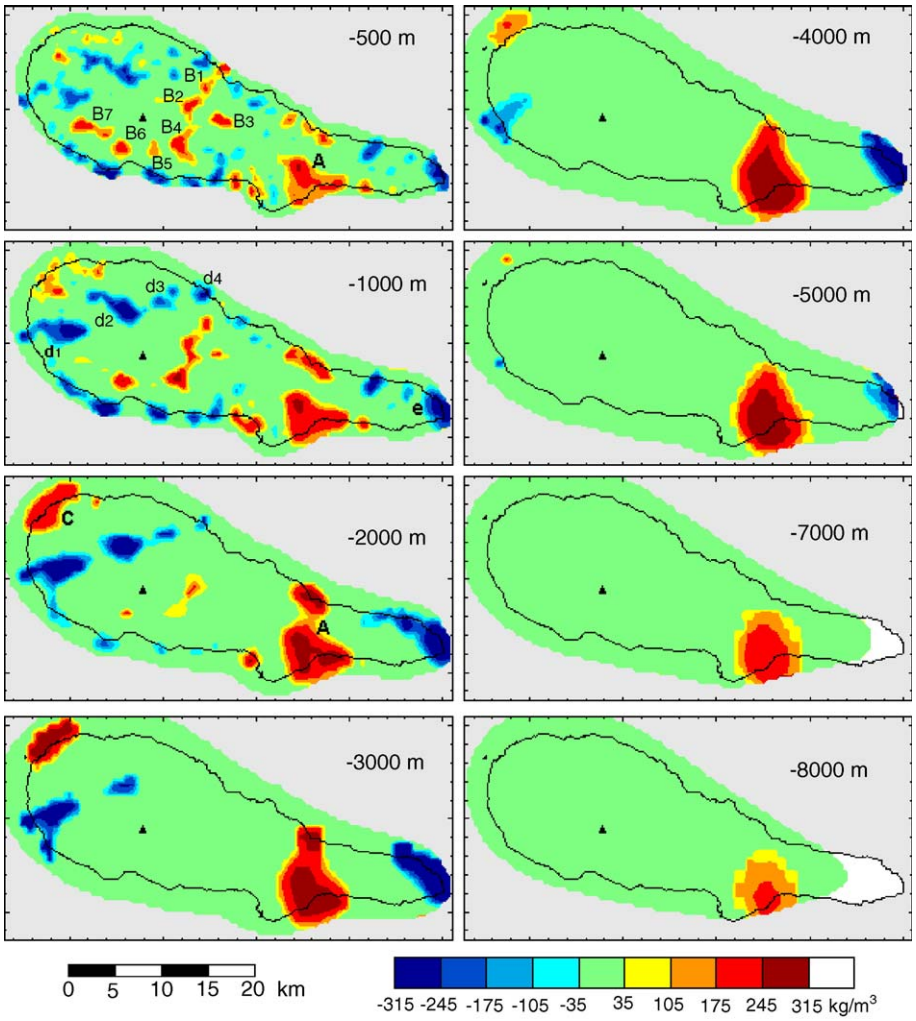

Fig. 7. Horizontal cross sections obtained from the gravity inversion model of anomalous density contrasts. Depth of the section (−500, −1000, …), in meters, is shown on top right. The triangle corresponds to Pico Mountain summit.

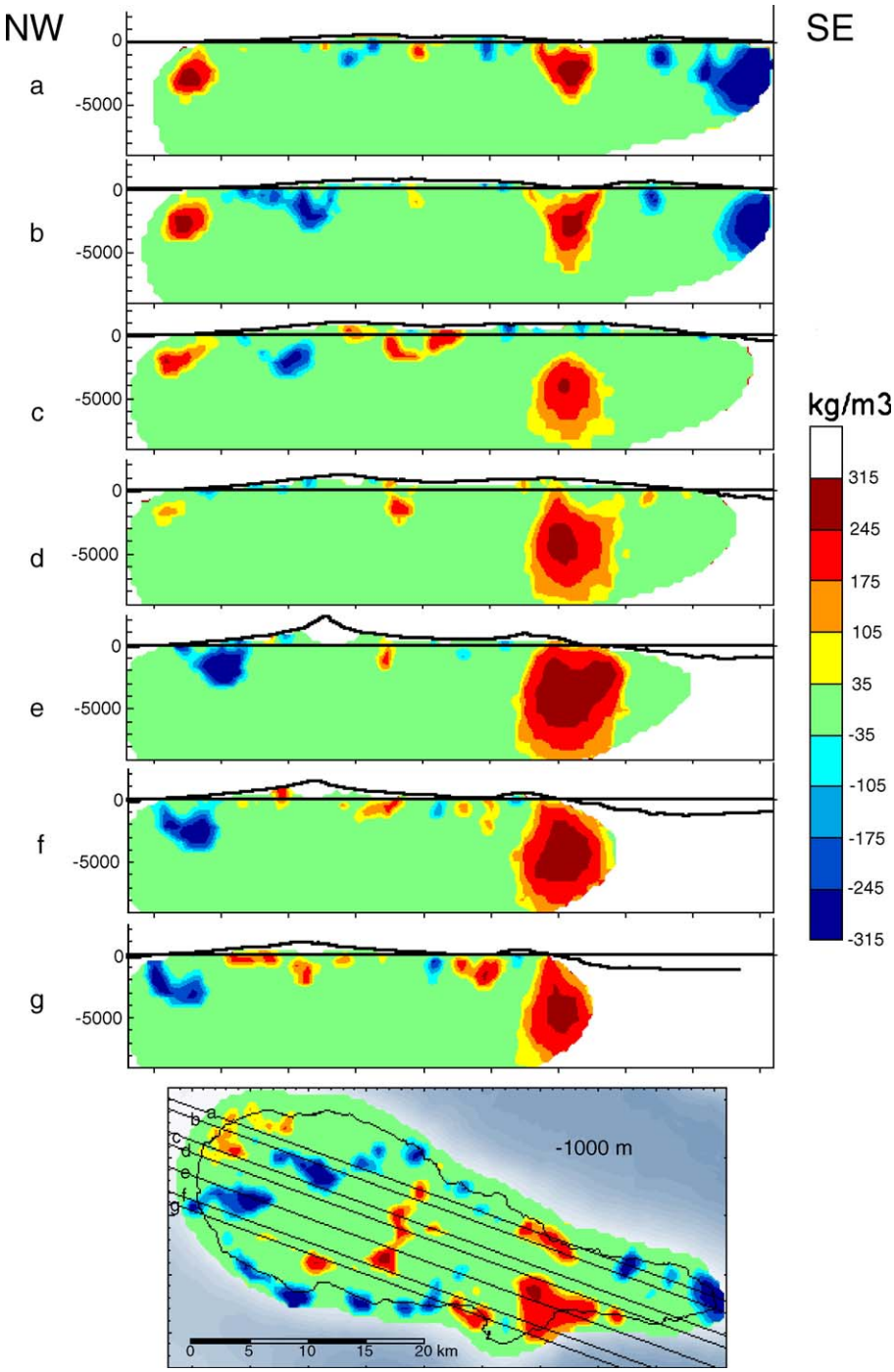

Fig. 8. Vertical profiles obtained from the gravity inversion model of anomalous density contrasts across several WNW–ESE lines.

beneath the centre of the island, including Pico Mountain; and (C) a body beneath the NW border of the island, close to Madalena, which at shallow depths is composed of several smaller bodies.

2. Most of the low-density bodies appear in peripheral locations with respect to the island, in part resulting from extrapolations and compensations of the model; therefore, this observation must be considered as equivocal. The low-density volumes located within the island are much smaller than the volumes of high-density bodies and include: (a) several intermediate-depth structures aligned on a general SW–NE trend

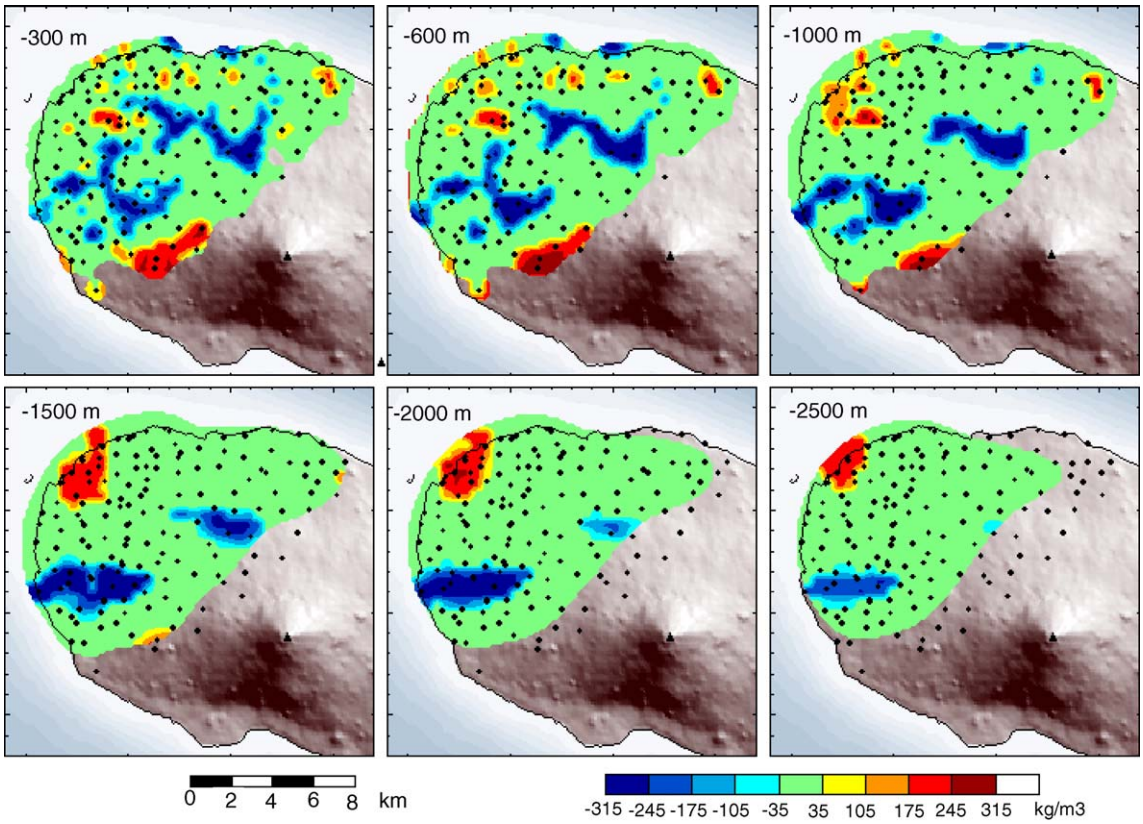

Fig. 9. Horizontal cross sections for the local model of the westernmost sector of Pico Island. Depth of the sections (−300, −600, −1000, ...), in meters, is shown on top left. The triangle corresponds to Pico Mountain summit.

in the western side of Pico Mountain; (b) a strong NW–SE-trending low in the eastern end of the island, near Ponta da Ilha; and (c) several very shallow and small lows scattered over the entire island.

3. The high-density body "A" located near Ribeiras, on the SE of the island (Fig. 7), clearly is the most prominent anomalous gravity structure found in Pico. It is a vertically and horizontally extended body that exhibits a nearly prismatic geometry, extending from the bottom of the model (8 km) to almost surface levels. Its horizontal configuration (see horizontal sections for mean depths) is well defined and clearly elongated along a NNW–SSE trend (about N160E). This trend is close to that of the Arrife Fault scarp (N10W to N20W — Figs. 3 and 4), which, facing East, is located on the SE border of the high-density body located near Ribeiras. This major body extends to the north, connecting with another body located near Praínha, in the NE coast of Pico (Fig. 7). The 3-D model yields a value of 12×10^{13} kg for the (additional or anomalous) mass in this area A, located at mean depth of 4.5 km and having a corresponding volume of 310 km^3.
4. The high-density structure "B" located in the central zone of the island close to Pico Mountain, has an irregular shape and is composed of several small and shallow nodules ("B_1", "B_2",...), that appear to be connected by narrow "bridges." The total (additional or anomalous) mass of these structures is 3.6×10^{13} kg at a mean depth of 1.3 km. These bodies do not appear to extend to depths greater than 2–2.5 km (Figs. 7 and 8). We also observe that the summit crater of Pico Mountain is located close to the centre of a semi-circular arc (with average diameter of 8 km), outlined by nodules "B_2", "B_4", "B_5", and "B_6"(see shallowest cross section).
5. A high-density body "C" located near Madalena which is predicted by the model adjustment, extends to greater depths (4–5 km) than the bodies ("B_1", "B_2",...) located beneath Pico Mountain (Fig. 7). That body extends to north of Madalena village and, because of its peripheral position, its size, shape and westward extension cannot be fully determined. In any case, body C occupies an extensive graben

structure (the Pedro Miguel Graben), best seen on the NE part of Faial Island and in the channel between Pico and Faial Islands.

6. An important zone of low-density bodies is located beneath the NW flanks of Pico Mountain. It consists of several discrete bodies ("d_1", "d_2", …), that all together define a linear structure with a N73°E azimuth (Fig. 7). This zone extends from Cais do Pico (on the NE) to Monte (on the SW coast) and downward to depths close to 4 km. A total volume of about 69 km^3 is estimated for these bodies, and, at shallower depths (see the −500 m section), they display elongated shapes along a N115°E azimuth. In the upper sections of the local model, to depths of about 600 m (Fig. 9), bodies "d_1" and "d_2" have a more widespread expression and seem to merge, defining a circular structure with an average diameter of 5 km.
7. A major low-density body appears present in the easternmost part of Pico Island, in the area of Ponta da Ilha (body "e" in Fig. 7), extending from the surface to about 5 km depth. Despite its peripheral position, this body of negative gravity contrast is well defined and larger on the deeper sections of the model, with a general NW–SE trend.
8. In the shallower sections of the model (to maximum depths of about 1 km), some lineaments of small negative density contrast bodies can be identified, mostly with WNW–ESE trends. In some cases, these lineaments can be traced along with high-density bodies, as it is the case for the lineament roughly located between São Miguel Arcanjo and Santo Amaro, along the NE coast of Pico Island. Less commonly, alignments with a general NE–SW trend can be found in the eastern part of Achada Plateau, for example close to Terra Alta (see also Fig. 4).

8. Discussion

The gravity-inversion method is highly sensitive to, and controlled by, the deep structures of the island (at more than 1.5 km depth). This is probably the reason why most of the surface geology of Pico Island is not easily reproduced by the gravity-inversion model, in particular the constructs of the Strombolian and fissure-vent volcanism of the Achada Plateau volcanic ridge. The same is true for the structures (to 1 km depth) related with the recent volcanism (as volcanic centres or eruptive fissures) and tectonics (as cone lineaments and fault scarps) that have been mapped (e.g. Fig. 4).

According to Rymer and Brown (1986), positive gravity anomalies in volcanic environments characterise mainly basaltic volcanoes and are caused by relatively dense sub-surface intrusions, which contrast with their surroundings either because those intrusive bodies are more mafic than surface lavas or, more likely, because near-surface eruptive products are generally uncompacted and more vesiculated. Ryan (1987 in: Malengreau et al., 1999) proposed that complexes of sub-volcanic intrusions and cumulates develop upwards during growth of a volcano. As a result, it is expected that column-like bodies of high-density rock would characterize the centre of a mature oceanic shield volcano. Closer to the surface, swarms of sheeted intrusions (dykes and sills) represent conduits for magma transport from the reservoirs to the surface or to shallow intrusions, the setting of this latter type of intrusions being well documented in active volcanoes, such as Kilauea or Piton de la Fournaise (Malengreau et al., 1999).

Moreover, with earlier gravity studies of the Azores islands, the positive anomalies generally correspond to large and deep intrusive bodies and dyke systems associated to old volcanic complexes, like in the Nordeste basaltic shield (São Miguel Island—Camacho et al., 1997). Subsidence phenomena are also commonly observed for these large positive bodies. Negative anomalies are often correlated either with the presence of faults and tectonic lineaments or are attributed to scoria deposits and other type of loose volcanic products, as is the case for the pyroclastic and alluvial infilling deposits of São Sebastião volcanic crater, on Terceira Island (Montesinos et al., 2003).

It has been proposed (Nunes, 1999, 2003) that Terras Chãs elongated depression (Fig. 3) was formed, about 30,000 to 35,000 years BP, by a lateral gravitational movement of the flank of Topo volcano. This flank-dislocation mechanism is an alternative to mechanisms involving a caldera collapse or landsliding of the shield volcano surface as proposed by others (Woodhall, 1974; Madeira, 1998). Driven by gravitational and tectonic instability, that lateral movement had occurred along deep, and curved faults (i.e., listric faults—Walker, 1988), like the Arrife Fault (Fig. 3), which borders the Terras Chãs depression on the west. Our data, namely the location and the depth of that gravity anomaly (that extends to about 8 km) seem to confirm the deep nature of the faults bordering Terras Chãs depression.

As mentioned previously, the most prominent gravity anomaly of Pico Island is the positive anomalous body located in Ribeiras and Terras Chãs area, on the SE slopes of Topo shield volcano, and according to Rymer

and Brown (1986), it is reasonable to infer that this positive anomaly reflects the presence of a major intrusive system beneath Topo volcano. This volcano is composed of about 90–95% of basaltic flows (Nunes, 1999), some of them very porphyritic (e.g., ankaramites), thus dense bodies, and the area also includes the older geological formations of Pico Island. In conclusion, the high-density body in the Ribeiras area may represent the "mature" core of a dense shield volcano and its deep roots, as well as the existence of deep (listric) faults in the area (e.g. Arrife Fault), namely bordering major collapse structures, well expressed topographically (see Figs. 3 and 4).

A similar conclusion can be drawn for the high-density body located near Madalena, as suggested by recent preliminary gravimetric studies done at Faial Island (Camacho and Vieira, 2002). From the available information, it is possible to infer the presence of an elongated positive density body under the NE sector of Faial Island, which extends SE to Madalena (Pico Island). This positive body, with a general WNW–ESE trend, might be related with the intrusive system emplaced in the Espalamaca–Ribeirinha–Salão shield volcano area (Faial Island) and it also delimits the Pedro Miguel Graben (Faial Island), a major volcano-tectonic depression with topographic and bathymetric expression in the channel between these two islands.

The WSW–ENE lineament of low-density bodies found in the northwestern flanks of Pico Mountain volcano has no direct correlation with any obvious surface volcanic or tectonic features in that sector of the stratovolcano (see Fig. 4). However, at shallower depths (see the −500 m section in Fig. 7), some of the discrete bodies display an elongated form along a N115°E azimuth. Even though the bad resolution of our models at shallower depths enables us to establish a more definite relation with the superficial structures, that trend is close to the dominant direction of the tectonic lineaments in the Azores archipelago and at Pico Island.

It is worthwhile mentioning that eruptive centres with WSW–ENE to SW–NE trends have been observed during historical volcanic eruptions in the Azores archipelago. Such was the case for the Capelinhos volcano submarine centres during the early phases of the 1957–1958 eruption (Machado, 1958) or the SW–NE alignment of the several eruptive loci during the 1998 Serreta eruption (Forjaz et al., 2001). Therefore, the anomalous low-density bodies, at least the shallower ones, could be attributed either to an extensional structure, or to a leaky transform structure with a general WSW–ENE trend, concealed by Holocene lava flows from Pico Mountain volcano. In this context, the intersection between this fracture zone and other sets of tectonic lineaments may facilitate magma intrusion from depths, constraining the location of the eruptive centres (Lourenço et al., 1998).

Machado (1956) considers the existence of a large magma reservoir centred beneath Pico Mountain, over which the volcanic edifice "floats" and is in isostatic equilibrium with the nearby "Planalto da Achada." From the analysis of anomalous isoseismal-line distribution and the characteristics of seismic signals, Machado and Forjaz (1965) and Machado et al. (1994) conclude that this reservoir, with a rough ellipsoidal shape, is at a mean depth of 5 km, has a thickness of 4.4 km and a diameter of 17.1 km (Fig. 10). Such a shallow magma system would represent an anomalous body beneath Pico Mountain volcano, with estimated volumes of about 520 to 670 km^3, depending on the shape of the reservoir.

The inferred presence of such a voluminous magma reservoir, however, is not compatible with geological, petrological and geochemical data collected in recent years (cf. Chovelon, 1982; Nunes, 1999; França, 2000). In fact, there is only a single evolved lava flow in Pico

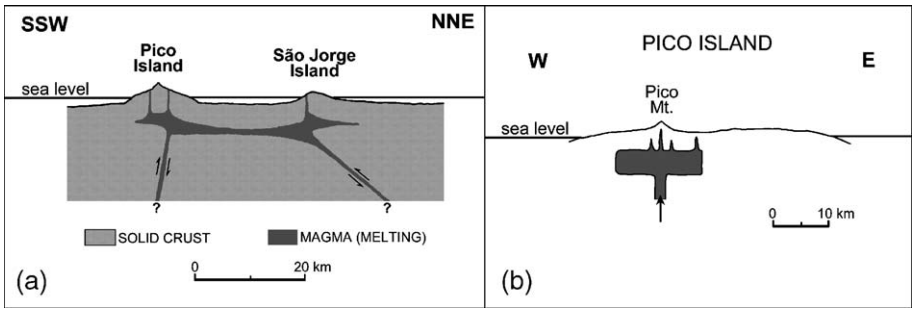

Fig. 10. Schematic cross sections of magma chamber inferred for Pico Island: (a) SSW–NNE profile (adapted from Machado and Forjaz, 1965) and (b) W–E profile (adapted from Machado et al., 1994). Results from gravity inversion (see Fig. 8) are not consistent with the sub-surface structures shown. See text for details.

Island, with a mugearite–benmoreite composition, which resulted from a process of fractional crystallization (at depths of about 20 km) of a portion of the magma intruded into the central feeding conduit of the Pico Mountain volcano (França, 2000). Additionally, the extrusion of magmas with distinctive mineralogical and textural features suggest that the eruptions from the Pico Mountain central vent(s) were not fed from a single large shallow reservoir (Chovelon, 1982) but, on the contrary, were fed directly from discrete sources at deeper levels in the crust or in the mantle.

It appears more likely that small magma pockets probably exist beneath Pico Mountain at shallow crustal levels; these pockets may ultimately interconnect to form some sort of larger reservoir, even though at present times they appear to be in a very early stage of development (França, 2000). Such smaller reservoirs can more consistently account for the results obtained from geological, petrological and geochemical studies mentioned above. Moreover, our gravity data (Fig. 8) appear to substantiate this hypothesis, as the existence of a huge anomalous body (of either positive- or negative-density contrasts) having the size and location shown in Fig. 10 beneath Pico Mountain cannot be confirmed. On the contrary, the major anomalous body in that area is the WSW–ENE string of low-density bodies along the northwestern flank of Pico Mountain, estimated to have a total volume of 69 km^3 and depths of 1 to 3 km.

An alternative interpretation of these discrete low-density bodies (Fig. 7) is that they correspond to complex intersections of different sets of tectonic lineaments (namely WSW–ENE and WNW–ESE trending fractures). Such intersections in turn create structurally weak zones, in which magma may have intruded into shallow depths beneath Pico Mountain. If so, these anomalous low-density bodies would indicate the presence of small reservoirs (10–20 km^3) beneath Pico Mountain (depths of 2–5 km) where magma may be stored. The same interpretation would be valid for the low-density body "e" located in the easternmost part of Pico Island (the Ponta da Ilha area), which extends to depths of about 5 km and corresponds to an area with geologically young volcanism (i.e., late Holocene).

The summit crater of Pico Mountain is located close to the centre of a semi-circular arc (8 km average diameter) delimited on its SE sector by shallow high-density bodies ("B" bodies). Based on the lack of gravity data on the higher slopes of Pico Mountain, we propose that this arcuate structure might outline a ring fracture associated with the collapse of the central sector of the edifice early in its growth. This inferred central collapse, however, has no present-day morphological expression, but Pico Mountain summit has collapsed at least twice in the past 5000 years, resulting in the formation of the summit crater, and an older crater partially preserved at 2050 m elevation (see Figs. 2 and 4).

9. Conclusions

This paper presents recent gravimetric data obtained for Pico Island, with a focus to better define its deep volcano-tectonic structures, the main conclusions being presented in previous sections. This study demonstrates the importance of gravimetric measurements in active volcanic areas as an important tool to delineate subsurface configurations of magma systems. Eruptive history of Pico Island shows that 14 eruptions have occurred in the past 1000 years and more than 35 eruptions in the last 2000 years, not evenly distributed in time and with longer repose times for the "Planalto da Achada" than on Pico Mountain volcano (Nunes, 1999).

Given that the average repose time of 130 years for the entire island is well overcome, and that Pico Mountain shows a higher eruptive frequency, we can anticipate that future eruptive activity will most likely occur at Pico Mountain volcano due to its higher eruption frequency. Renewed activity could involve Hawaiian to Strombolian eruptions, either located at the summit, or as flank eruptions on the western or south-eastern slopes of the Pico Mountain. Based on the results obtained from the gravity surveys in assessing the potential volcanic hazards on Pico Island (e.g., extrusion of lava flows — França et al., 1999), we recommend that volcano-monitoring programmes for Pico Island place high priority on the area beneath which the low-density bodies are inferred, where small, sub-surface magma reservoirs may be forming. The same applies for the Ponta da Ilha area, where a large low-density body has been identified. This area has been the site of prehistoric lava extrusions, a few centuries before the settlement of Pico Island (Nunes, 1999). Here, microgravity monitoring may help tracking magma migration, variations in magma input, and changes in magma density. Such data in turn could play an important role in predicting eruptive events (Rymer and Brown, 1986; Rymer, 1994).

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